

OPIM639 · Innovations & Operations of Intelligent Online Marketplaces

Chagee: opening overseas stores right

Predict demand, provision the store, size the counter

Group [TBC]

[Member names TBC - confirm with the team]



Chagee's supply chain is fixed, so new-market risk lands on how much a store opens with and how fast its counter serves



Scaling fast into markets it has not run before

7,453 stores at end-2025, 345 of them overseas in 7 markets, and about 200 more overseas openings planned for 2026. FY2025 GMV RMB 31.58bn, net revenue RMB 12.91bn.



The upstream is rigid by design

Chagee says it sources tea and milk directly for domestic operations, with distribution run from HQ. Reported own Yunnan plantations and a narrow menu (tea lattes about 91% of China sales, secondary source) keep sourcing fixed.



New-store forecasts lean on local judgement

Our reading: a new market's opening forecast is judgement-led by a local General Manager, while service is moving toward semi-automation (QR-scan dispensing, Chagee Express).



Why not a marketplace lens

Chagee runs one closed channel, app to store: no third-party sellers, no sell-vs-FBA choice and no online-versus-offline conflict. We leave the marketplace-design topics aside; the franchise GMV revenue share is the one overlap, and it is out of scope.

● INTERACTIVE EXPLAINER

WHERE DOES UNCERTAINTY LAND?

Our reading of public sources, not a Chagee statement. With sourcing fixed, a demand miss can only be absorbed in the store: by the stock it opened with, or by how fast the counter serves. Local sourcing would ease stock risk, but the counter stays exposed either way.

Rigid sourcing

Localised sourcing

Rigid sourcing Chagee today, our reading

Sourcing ● ● ● Fixed: single origin

Distribution ● ● ● Fixed: run from HQ

Opening stock ● ● ● Absorbs the miss

Counter speed ● ● ● Absorbs the miss

Localised sourcing For contrast only

Sourcing ● ● ● Flexes with local suppliers

Distribution ● ● ● Shorter, local

Opening stock ● ● ● Less exposed

Counter speed ● ● ● Still exposed

● ● ● low ● ● ● high exposure to a demand miss

Printed with both columns shown. Live: the toggle highlights one.

OUR VIEW

Take the rigid upstream as given; fix the store-level decisions.

New-market uncertainty lands on three decisions we can model: the demand forecast (slide 2), the opening stock (slide 3) and the counter setup (slide 4).

1 · Why this slice

About 1.5 minutes. Start with scale: Chagee is opening overseas faster than it can learn each market. Then the key move: Chagee's upstream is rigid by design. It sources tea and milk directly and runs distribution from HQ; reports describe its own Yunnan plantations and a menu dominated by tea lattes. That is a strength, but it means sourcing cannot soak up a demand miss. So where does the miss go? Into the store: the stock it opened with, and the speed of its counter. Use the toggle: with localised sourcing, stock risk eases, but the counter stays exposed either way. Say clearly that this is our reading, not a Chagee statement. On the marketplace weeks: Chagee has one closed app-to-store channel, so seller and channel-conflict topics do not apply; the franchise revenue share is the one overlap and is out of scope.

Sources: Scale facts: L-jiemian. Direct sourcing and HQ distribution: L-prn (company release). Yunnan plantations and tea-latte share: L-prometheus (SECONDARY). Narrow menu choice: L-chainreact (secondary). Semi-automation: L-unsw, L-cyberrrt. GM-led forecasting: I-leeyen (INFORMAL – check with Lee Yen before presenting).

App-only ordering lets Chagee read a new market’s demand before the store opens, and tell opening hype from repeat habit

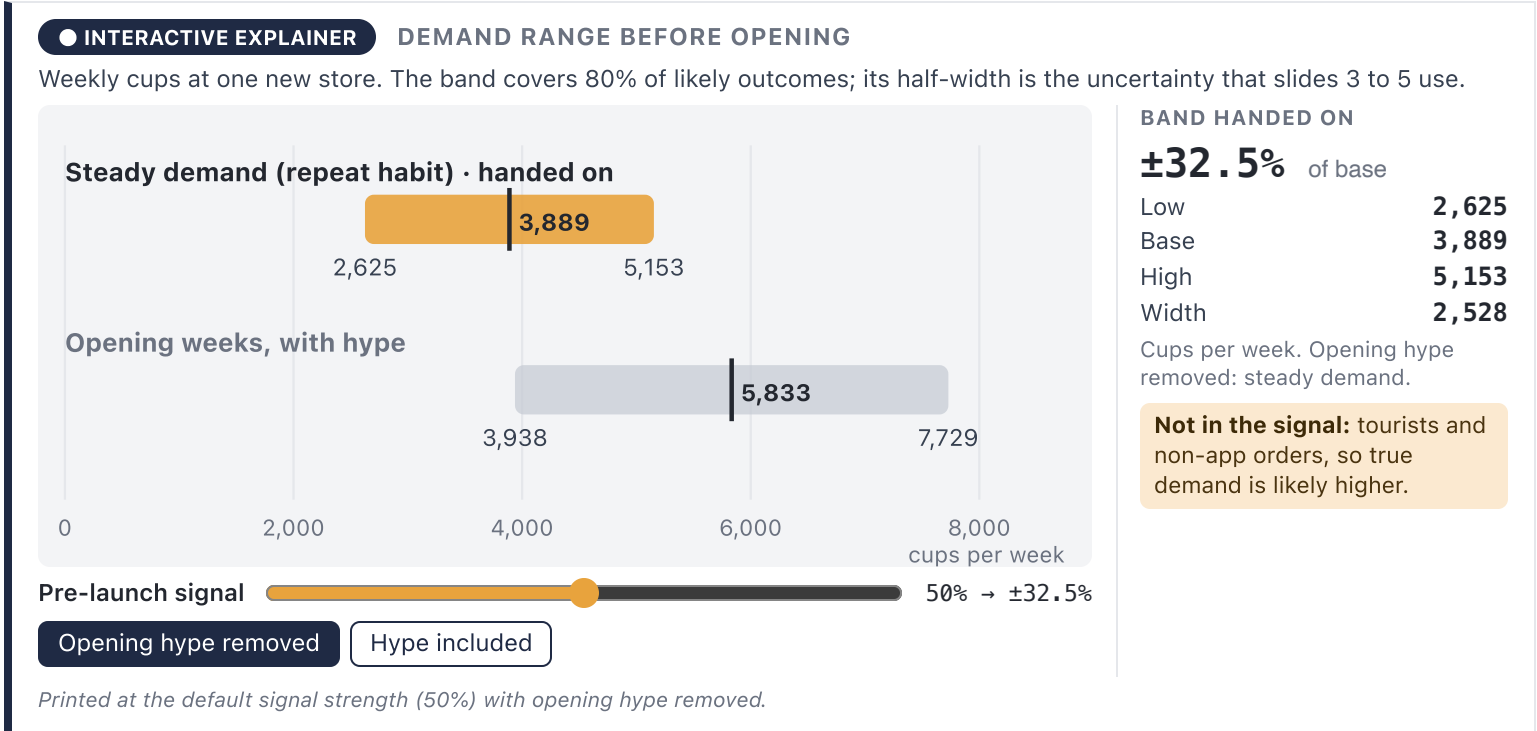
- Every order is tied to a member**

With app-only ordering (our understanding), each cup is linked to a member account, so a pre-launch waitlist is a demand signal, not a guess.
- More signal, a narrower band**

The forecast is a range, not a number. As sign-ups build before launch, the band narrows from ±50% to ±15% of the base (illustrative).
- Hype versus habit**

Opening crowds are not steady demand: the Kuala Lumpur store reportedly sold 1,000+ cups on its first day (secondary source). Member IDs separate first orders from repeat orders.
- Blind spot: tourists and non-app orders**

Visitors and tourist exceptions are outside the member signal, so the forecast understates demand where tourists matter. We name this gap rather than model it.



2 · Predict

About 1.5 minutes. Because Chagee orders go through the app (our understanding from an industry conversation, to be confirmed), every cup is tied to a member. That turns a pre-launch waitlist into a demand signal. Move the slider: with no signal the band is wide; as sign-ups build it narrows. The width of this band is the uncertainty we carry into the next three slides. Toggle the hype: opening crowds, like the reported opening day in Kuala Lumpur, are not steady demand, and member IDs let us separate first orders from repeat orders. Be upfront about the blind spot: tourists and non-app orders are invisible to this signal, so the forecast understates demand in tourist areas. Novelty: AI forecasting and siting exist, e.g. Starbucks Deep Brew; our point is using a closed app channel to read demand before the store exists, to our knowledge.

Sources: App-only ordering with tourist exceptions: I-leeyen (INFORMAL - check before presenting). Kuala Lumpur opening day: L-gabgrowth (secondary). Not novel - AI forecasting and siting: L-starbucks.

The wider the demand range, the costlier the opening-stock decision, especially with fresh milk



Two ways to be wrong

Too much fresh milk expires; too little turns customers away. Week 1 newsvendor logic: stock up to the point where the next litre is as likely to be wasted as to save a sale.



Here a stockout costs more than waste

A missed cup loses S\$3.82; a spare litre of milk wastes S\$2.55. So the best order sits high in the band, at the 91% point of demand.

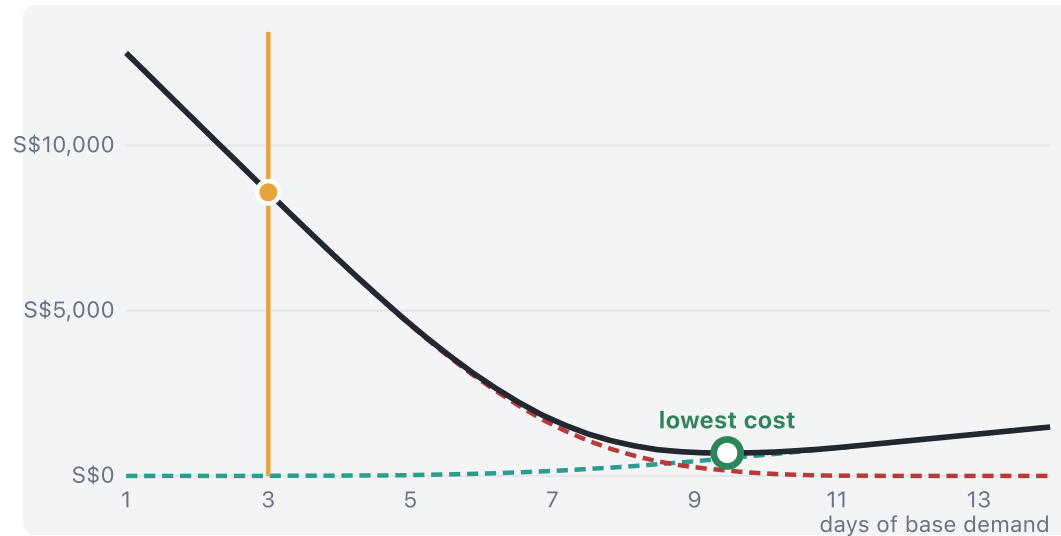


Shelf life caps the buffer

Fresh milk keeps about 7 days (illustrative), so extra stock cannot be carried forward. A narrower band from slide 2 is the cheaper fix.

● INTERACTIVE EXPLAINER OPENING STOCK CALCULATOR

Expected cost of the first fresh-milk order at one store, using slide 2's band. Overseas milk sourcing is unknown; figures are generic.



■ Total expected cost ■ Milk wasted ■ Cups missed

Days of opening stock 3 days

Printed at the band from slide 2 and the default of 3 days of stock.

AT 3 DAYS OF STOCK

S\$8,581 expected

Milk wasted **S\$3**

Cups missed **S\$8,578**

BEST ORDER AT THIS BAND (±32.5%)

9.5 days **S\$694**

Same order, strong signal **S\$320**

Hand-off: open with 788 L of milk. Counter (slide 4): cheapest is All-human; mixed beats full automation by S\$0.25/cup.

OUR VIEW

Narrow the band before you stock the store.

From no signal to the default signal, the expected cost of the best opening order falls from S\$1,064 to S\$694 per store (illustrative). Hand-off to slide 4: the same band sizes the counter.

3 · Provision

About 1 minute. This is Week 1 newsvendor logic applied to fresh milk. Too much expires, too little loses sales, and here a lost cup costs more than a wasted litre, so the best order sits high in the band. Show the curve: the slider is days of opening stock; the dot marks the cheapest level. Then go back to slide 2 in your head: a wider band pushes the best order up and makes even the best decision more expensive. That is the value of the pre-launch signal in stock terms. Caveats: we assume the first order covers one shelf life before regular deliveries, and milk costs are generic because Chagee's overseas milk sourcing is not public.

Sources: All figures: assumptions.json (illustrative). Overseas milk sourcing: unknown, deliberately not claimed (brief.md section 9).

At a new store's volume, a mixed people-and-machine counter beats full automation, because machines sit idle, not because demand is uncertain

We tested the hypothesis that mixed wins only until demand becomes predictable. Our simulator rejected the uncertainty part.



Machines come in whole units, so they sit idle

Even one machine per station serves about 240 cups an hour against a forecast peak of about 61. The fully automated line is busy only 21% to 22% of open hours.



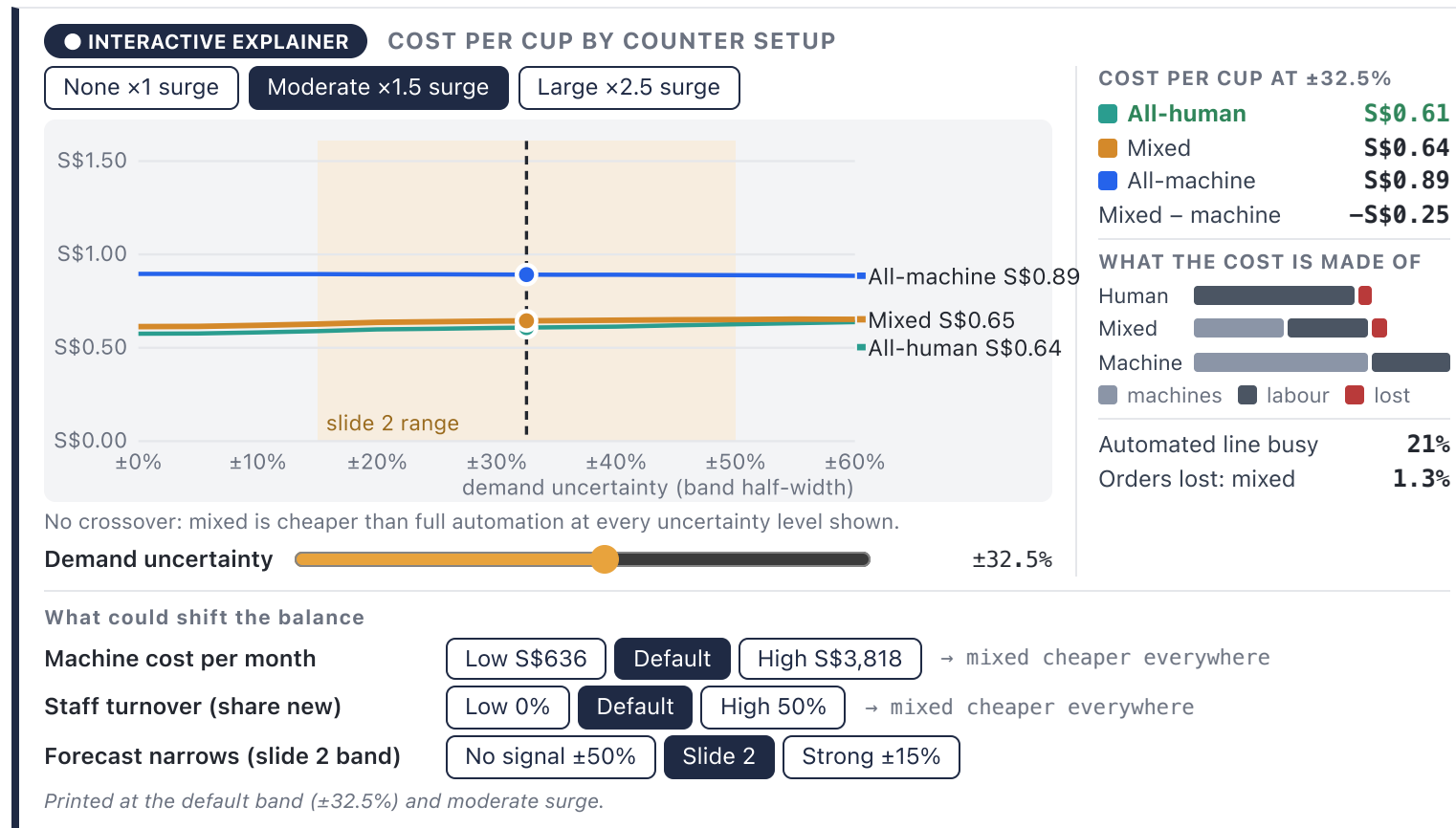
People flex; machines cannot be returned

Rosters follow the hour and the real demand, and staff rebalance across stations like a bucket brigade. A mixed counter buys 3 machines instead of 6, with similar staff hours.



Where machines still win

Big surges: at a 2.5x peak, people-based counters lose about 11% of orders and full automation is cheaper. Also steady high volume, consistency and labour scarcity.



OUR VIEW

Start lean and mixed; automate once volume and peaks are proven, not just once the forecast narrows.

In our model an all-human counter is cheaper still (\$0.61 vs \$0.64 per cup at the default band), but it is the most exposed to surges and turnover. Machine breakdowns are not modelled, which flatters automation.

4 · Perform

About 2 minutes, the centrepiece. Our hypothesis was: a mixed counter beats full automation until demand becomes predictable. We built a simulator to test it, and it only half holds. Mixed does beat full automation at the default settings, at every level of uncertainty. But the reason is not uncertainty. Machines come in whole units: one per station already has several times the peak demand of a new store, so the automated line sits idle most of the day and its fixed cost is spread over too few cups. Move the uncertainty slider: the lines barely move. Switch to a large surge: now the people-based counters overflow and full automation wins. So the real trigger for automation is proven volume and peaks. Be honest about limits: all-human is cheapest in our model; the pooled staffing formula slightly overstates a real bucket brigade (our Week 5 check missed its tolerance, gap 4.7 to 6.8 percent); breakdowns are not modelled. To our knowledge, simulating a store before it exists, fed by its demand forecast, is our contribution; store digital twins today replicate existing stores.

Sources: All figures: sim/ with assumptions.json (illustrative). Semi-automation context: L-unsw, L-cybertr. Not novel - beverage automation: L-gongcha (company-reported), L-slate; store digital twins: L-mosimtec. Labour cost floor: L-mom-pwm, L-cpf.

Even modest per-store savings compound across about 200 overseas openings a year, if the risks are managed

● INTERACTIVE EXPLAINER BENEFIT CALCULATOR

A sizing of the prize, not a forecast: first-year, new stores only, each term an assumption you can move.



Linked: band $\pm 32.5\%$ from slide 2, moderate surge.

Re-link to slides 2–4

≈ S\$10.0m

a year, illustrative

200 openings × (200,000 cups × S\$0.25 saved per cup + S\$370 stock cost avoided)

≈ S\$9.9m from the counter + S\$0.1m from opening stock. Saving per cup = full-automation minus mixed cost per cup; stock cost avoided = best-order cost with no signal minus with the slide 2 band.

Printed at the linked defaults: saving per cup from slide 4, waste avoided from slide 3.

OUR CONCLUSION

The saving per cup is small; the case is the count of openings.

Our simulator rejected the claim we set out to prove; what survives is simpler: start lean and mixed, read demand before opening, and automate on proven volume. To our knowledge, based on public sources, simulating a store before it exists is the new part.

RISKS

1 Staff dependence and turnover

A people-based counter relies on trained staff, and new hires are slower. Keep SOP-led training and the flexible worker role; track turnover as in slide 4.

2 Tourist and non-app demand is invisible

The member signal misses visitors, so the band sits too low in tourist areas. Widen it there and log walk-up exceptions after opening.

3 Opening crowds read as steady demand

Sizing to opening-week queues overbuys machines and staff. Separate first orders from repeat orders before committing capex.

4 Automation versus the premium teahouse

Heavy automation could undercut the premium experience. The mixed setup keeps people at the customer-facing steps: finishing and handover.

Next step with real Chagee data

- Replace every assumption with store actuals: station times, wages, machine costs.
- Back-test the waitlist signal on recent openings, such as Korea (launch planned for Q2 2026).
- Pilot mixed versus automated counters in paired new stores.

5 · Benefit, risks, next step

About 1 minute. The benefit is a formula, not a forecast: stores opened, times cups per store, times the saving per cup from slide 4, plus the stock cost avoided from slide 3. Each term is a slider; the defaults are linked to the earlier slides. The per-cup saving is cents, but it repeats across about 200 overseas openings a year. Then the risks: click through each. Staff turnover; tourists invisible to the app signal; opening crowds mistaken for steady demand; and automation versus the premium teahouse feel. Close with the next step: with Chagee's real data we would replace every assumption, back-test the waitlist signal on recent openings such as Korea, and pilot mixed against automated counters.

Sources: Overseas openings planned for 2026 and Korea launch: L-jiemian. Everything else: assumptions.json and sim/ (illustrative).